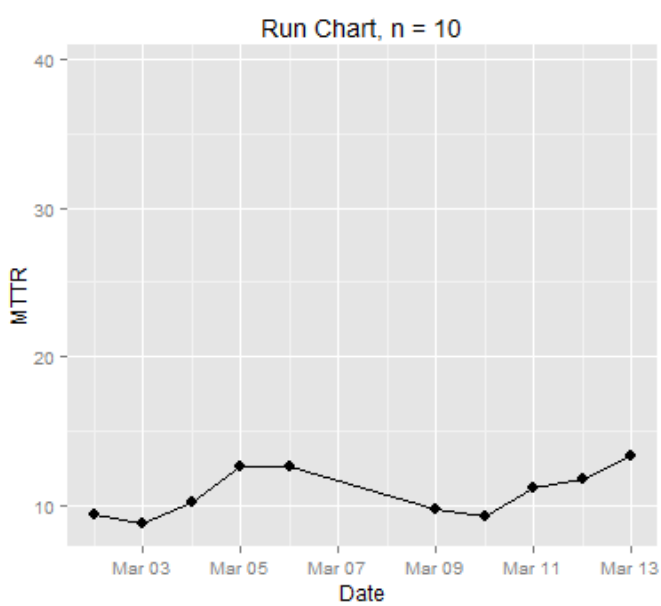
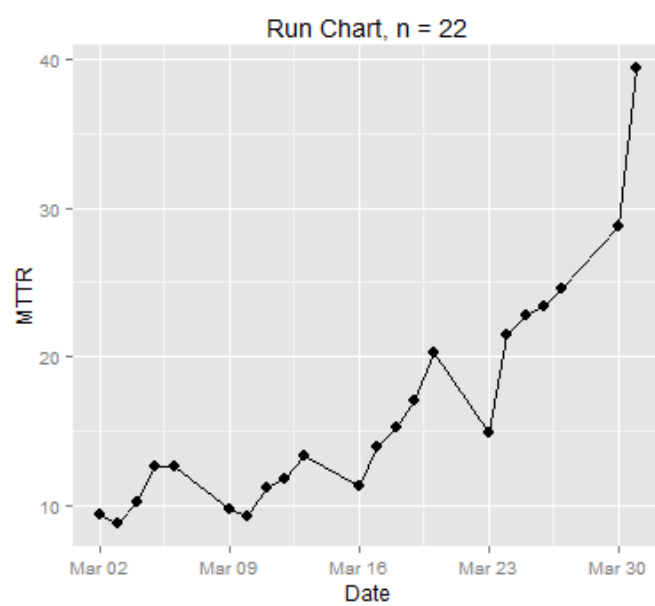
The data set consists of tabulated MTTR (mean time to resolve) tickets on a daily basis where weekends have been omitted. Only the MTTR column is used for calculating/testing ranks. Test of rank is used because it is more robust for time-series data. The chosen level of significance for both tests is 0.05.

			n = 10			n > 10		
Date	Tickets	MTTR	Rank	$\sum_{i=1}^{n-1}[rank(X_i)-(X_{i+1})]^2$	$\sum_{i=1}^{n}\left[rank(X_i)-\frac{(n+1)}{2}\right]^2$	Rank	$\sum_{i=1}^{n-1}[rank(X_i)-(X_{i+1})]^2$	$\sum_{i=1}^{n}\left[rank(X_i)-\frac{(n+1)}{2}\right]^2$
Mar 2	3218	9.4	3	4	6.25	3	4	72.25
Mar 3	2966	8.7	1	16	20.25	1	16	110.25
Mar 4	2759	10.2	5	12.25	0.25	5	20.25	42.25
Mar 5	2611	12.6	8.5	0	9	9.5	0	4
Mar 6	2355	12.6	8.5	20.25	9	9.5	30.25	4
Mar 9	2976	9.7	4	4	2.25	4	4	56.25
Mar 10	2727	9.2	2	16	12.25	2	16	90.25
Mar 11	2759	11.2	6	1	0.25	6	4	30.25
Mar 12	2811	11.8	7	9	2.25	8	9	12.25
Mar 13	2144	13.3	10	-	20.25	11	16	0.25
Mar 16	2937	11.3				7	25	20.25
Mar 17	2853	13.9				12	4	0.25
Mar 18	2656	15.2				14	1	6.25
Mar 19	2743	17				15	1	12.25
Mar 20	2285	20.2				16	9	20.25
Mar 23	3128	14.8				13	16	2.25
Mar 24	2752	21.5				17	1	30.25
Mar 25	2638	22.8				18	1	42.25
Mar 26	2294	23.4				19	1	56.25
Mar 27	1912	24.6				20	1	72.25
Mar 30	2925	28.8				21	1	90.25
Mar 31	2726	39.4				22	-	110.25





Exact Test for Small Sample, n=10

$$NM = \sum_{i=1}^{n-1} [rank(X_i) - (X_{i+1})]^2$$
$$NM = 82.5$$

$$RVN = \frac{\sum_{i=1}^{n-1} [rank(X_i) - (X_{i+1})]^2}{\sum_{i=1}^{n} \left[rank(X_i) - \frac{(n+1)}{2}\right]^2}$$
$$RVN = \frac{82.5}{82}$$
$$RVN = 1.006$$

H_O: The sequence of MTTR is random
H_A: The sequence of MTTR follows a trend

$NM = 82.5 \leq NM_{\alpha(0.05)} = 85$
$P[NM \leq 82.5 \mid H_O \text{ is true}] > 0.0260$

Based on exact test on both critical and p-value methods, the null hypothesis is rejected; there is significant trend.

Beta Approximation for RVN, n=22

From computation above of $RVN = 1.006$

H_O: The sequence of MTTR is random
H_A: The sequence of MTTR follows a trend

$RVN_{obs} = 1.006 \leq RVN_{\alpha(0.05)} = 1.32$
$P[RVN \leq 1.006 \mid H_O \text{ is true}] > 0.005$

Based on beta approximation test on both critical and p-value methods, the null hypothesis is rejected; there is significant trend.

Normal Approximation w/o Continuity Correction, n=22

$$NM = \sum_{i=1}^{n-1} [rank(X_i) - (X_{i+1})]^2$$
$$NM = 180.5$$

$$RVN = \frac{\sum_{i=1}^{n-1} [rank(X_i) - (X_{i+1})]^2}{\sum_{i=1}^{n} \left[rank(X_i) - \frac{(n+1)}{2}\right]^2}$$
$$RVN = \frac{180.5}{885}$$
$$RVN = 0.2040$$

$$E[RVN] = 2$$

$$Var = \frac{4(n-2)(5n^2 - 2n - 9)}{5n(n+1)(n-1)^2}$$
$$Var = \frac{4(22-2)[5(22^2) - 2(22) - 9)]}{5(22)(22+1)(22-1)^2}$$
$$Var = 0.1698$$

$$RVN \sim N(\mu = 2, \sigma^2 = 0.1698)$$
$$Z = \frac{0.2040 - 2}{\sqrt{0.1698}} = -4.360$$

H_O: The sequence of MTTR is random
H_A: The sequence of MTTR follows a trend

$Z = -4.360 \leq Z_\alpha = -1.645$
$P[Z \leq -4.360 \mid H_O \text{ is true}] < 0.0002$

Based on normal approximation using asymptotic distribution without continuity correction, null hypothesis is rejected; there is significant trend. This also supports the exact test using RVN statistic.

```r
# Code block to install packages
is.installed <- function(x) {
    is.element(x, installed.packages()[,1])
}

# Uses randtests package instead of lawstat
ifelse(!is.installed('randtests'), install.packages('randtests'), require(randtests))

# Since the dataset is too long to be in a vector, I prepared a CSV file in a dataframe format instead
url <- 'https://raw.githubusercontent.com/foxyreign/MasterOfStatistics/master/Stat%20266/Stat%20266%20-%20Ass
ignment%201%20(Runs%20and%20Ups%20and%20Downs).csv'
download.file(url, "MTTR.csv")
assign1 <- read.csv('MTTR.csv', head = T, sep = ',')
head(assign1)

##         Date Tickets MTTR
## 1 2015-03-02    3218  9.4
## 2 2015-03-03    2966  8.7
## 3 2015-03-04    2759 10.2
## 4 2015-03-05    2611 12.6
## 5 2015-03-06    2355 12.6
## 6 2015-03-09    2976  9.7

assign1$Date <- as.Date(assign1$Date, format = '%Y-%m-%d') # Converts to datetime series format
assign1$Subset <- ifelse(assign1$Date < '2015-03-13', 'Exact Test', 'Normal Approx')
mttr.range <- range(assign1$MTTR)

# Normal approximation w/o Continuity Correction, n = 22
bartels.rank.test(assign1$MTTR, alternative = 'left.sided', pvalue = 'normal')

##
##  Bartels Ratio Test
##
## data:  assign1$MTTR
## statistic = -4.3597, n = 22, p-value = 6.513e-06
## alternative hypothesis: trend

# Use exact calculation or beta distribution, n = 10
bartels.rank.test(assign1[1:10,3], alternative = 'left.sided', pvalue = 'auto')

##
##  Bartels Ratio Test
##
## data:  assign1[1:10, 3]
## statistic = -1.7088, n = 10, p-value = 0.04334
## alternative hypothesis: trend
```